

Annotated Bibliography — Assessment #3

Robin Kim · ISM I · Baseball Analytics Research

Reference 1

Scarborough, D. M., Colón, P. E., Linderman, S. E., & Berkson, E. M. (2021). Comparison of kinematic sequences during curveball and fastball baseball pitches. *Frontiers in Sports and Active Living*, 3, Article 699251. <https://doi.org/10.3389/fspor.2021.699251>

Using 3D motion capture on 14 pitchers who threw both curveballs and fastballs, this study found that nearly half of all pitches showed an altered “kinematic sequence” linked to higher elbow and shoulder strain. Orthopedic surgeons at Massachusetts General Hospital and Harvard Medical School conducted the work, and it discloses no conflict of interest. Where it thins out is sample size: only 4 of the 14 pitchers were high schoolers, and the paper never isolates results by level, so how often HS arms specifically hit that risky pattern is still an open question. Even so, I'll draw on this to argue that an MLB-informed injury model can miss risk factors already sitting in HS pitchers.

Reference 2

Mair, H., Kenney, N. A., Uhl, T. L., Ullery, L. R., & Hosey, R. G. (2022). Effect of pitching restrictions and mound distance on youth baseball pitch counts. *Orthopaedic Journal of Sports Medicine*, 10(7), Article 23259671221110547. <https://doi.org/10.1177/23259671221110547>

Comparing three youth baseball leagues, some capping pitch counts, others capping innings, with different mound distances, this study found that inning-limited leagues let pitchers throw significantly more per outing than their pitch-limited counterparts. Its authors are sports medicine physicians and athletic trainers at the University of Kentucky Medical Center, though one discloses a funding-related conflict of interest worth flagging rather than assuming full independence. The comparison itself is thin: just three leagues in a single city, so the pattern may not hold elsewhere. Still, it's useful evidence that copying an MLB-style restriction rule isn't sufficient by itself, since the two pitch-limited leagues here followed drastically different actual rules.

Reference 3

Kriz, P. K., Collins, C. L., Staffa, S. J., Zurakowski, D., Zaremski, J. L., Bowers, R. L., Kriz, J. P., & Sugimoto, D. (2026). Do pitching restriction policies reduce shoulder and elbow injuries in high school baseball players? *Orthopaedic Journal of Sports Medicine*, 14(2), Article 23259671251412404. <https://doi.org/10.1177/23259671251412404>

Does a national pitching-restriction mandate actually protect young arms? This study says not by itself: the NFHS's 2016 policy, built on MLB's Pitch Smart guidelines, showed little measurable effect on shoulder and elbow injuries nationwide, even though specific pieces of it, like mandatory rest days, did correlate with fewer injuries. The lead author is a sports medicine physician at Boston Children's Hospital, and the group discloses no conflicts of interest. Drawing on a large national injury database gives the study real scope, though it can only show correlation, not proof those components caused the drop. This is the piece I keep coming back to for my thesis, the clearest case of an MLB-derived

metric failing wholesale yet working once broken into HS-specific pieces.

Reference 4

Matsuura, T., Takata, Y., Iwame, T., Iwase, J., Yokoyama, K., Takao, S., Nishio, S., Arisawa, K., & Sairyo, K. (2021). Limiting the pitch count in youth baseball pitchers decreases elbow pain. *Orthopaedic Journal of Sports Medicine*, 9(3), Article 2325967121989108. <https://doi.org/10.1177/2325967121989108>

In 2018, one Japanese region swapped an innings-based pitch limit for a pitch-count-based one, and this study tracks what changed. Youth pitchers threw fewer pitches and reported significantly less elbow pain afterward. The authors, orthopedic researchers at Tokushima University, back this with a large sample and formal statistical testing, and the paper explicitly reports no conflicts of interest. The catch: these pitchers were 8 to 12 years old, not high-school age, and the design tracks one region before and after a single rule change rather than comparing multiple leagues side by side. I'll lean on this to back up the previous source's point: MLB-informed rules need to be fitted to the specific league and age group, not applied as a blanket standard.

Reference 5

Thomay, C. D. (2014). *Markov chain theory with applications to baseball* [Independent study thesis, College of Wooster]. OpenWorks. <https://openworks.wooster.edu/independentstudy/5722/>

Unlike the journal articles here, this source is an undergraduate math thesis reviewed only by a faculty advisor, not peer-reviewed, worth naming given the assignment's requirements. It replicates the sabermetric run-expectancy calculation normally built on MLB data, applying it instead to 2014 College of Wooster game data, and finds lower predicted run gains from bunts and steals given the team's own lower success rates. Because it is one Division III team's one season, the finding is illustrative rather than generalizable. That said, since it is a math method rather than time-sensitive stats, the core conclusion, that sabermetric run-expectancy formulas require level-specific input data rather than borrowed MLB averages, holds up just as well in 2026 as it did in 2014. I'll use it to show sabermetric formulas travel across levels, but only if the data plugged in matches the level being studied.